

HDFS Replication

1. What is HDFS?

HDFS = Hadoop Distributed File System

It is the distributed storage layer of Apache Hadoop.

Designed for:

None

Huge files

Big data analytics

Sequential reads/writes

Fault tolerance

Commodity hardware clusters

2. Core Components

A. NameNode (Master Metadata Node)

Stores metadata:

None

File names

Directory tree

Permissions

Which blocks are stored on which DataNodes

Replication metadata

Does **not** store actual file data.

B. Secondary / Standby NameNode

Used for:

None

Checkpointing metadata

Failover (HA setups)

Backup assistance

Modern HA uses Active + Standby NameNode.

C. DataNodes

Store actual file blocks.

Responsible for:

None

Read requests

Write requests

Block replication

Heartbeats to NameNode

Block reports

3. How HDFS Stores Files

Large file is split into blocks.

Example:

None

500 MB file

Block size = 128 MB

=> split into multiple blocks

Each block stored across different DataNodes.

Older defaults were 64 MB, modern clusters often use 128 MB or 256 MB.

4. What is HDFS Replication?

Each block is copied to multiple machines.

Example:

None

Replication Factor = 3

Block A stored on:

Node1

Node4

Node8

So if one node fails, data still available.

5. Why Replication?

Provides:

None

Fault tolerance

High availability

Parallel reads

Rack-level protection

Hardware failure recovery

6. Rack Awareness (Very Important)

HDFS avoids storing all replicas on same rack.

Typical placement:

None

Replica 1 → Local rack

Replica 2 → Different rack

Replica 3 → Same remote rack

Why?

If entire rack/network switch fails, data survives.

7. Write Flow in HDFS

Suppose client uploads file.

Step 1

Client asks NameNode:

None

Where should block go?

Step 2

NameNode returns DataNodes.

Step 3

Client streams block to first DataNode.

Step 4

Pipeline replication:

None

Client → DN1 → DN2 → DN3

Efficient network usage.

8. Read Flow in HDFS

Step 1

Client asks NameNode:

None

Where are replicas?

Step 2

NameNode returns locations.

Step 3

Client reads nearest DataNode replica.

Locality improves speed.

9. What Happens If DataNode Fails?

NameNode detects missed heartbeats.

Then:

None

Mark node dead

Use remaining replicas

Create new replica elsewhere

Automatic self-healing.

10. What Happens If NameNode Fails?

Historically single point of failure.

Modern solution:

None

Active NameNode

Standby NameNode

Automatic failover using ZooKeeper

Often integrated with Apache ZooKeeper.

11. HDFS Replication vs Database Replication

Feature	HDFS	Database
Data Type	Large files	Rows / records
Write Pattern	Append-heavy	CRUD
Replication Unit	Block	Row / page / log

Latency	High	Lower
Use Case	Analytics	OLTP

12. Why HDFS is Append-Only

HDFS optimized for:

None

Write once

Read many

Append logs

Batch analytics

Not ideal for frequent UPDATE / DELETE.

Reason:

- Simpler consistency model
- Easier replication
- High throughput

13. HDFS + Hive Partitioning

Apache Hive stores tables as directories/files.

Example:

SQL ▼

```
sales/date=2025-03-28/country=IN/
```

```
sales/date=2025-03-28/country=US/
```

Benefits:

None

Skip unnecessary files

Fast filtering

Scalable analytics

14. Small Files Problem

HDFS is bad at millions of tiny files.

Why?

Each file metadata consumes NameNode memory.

Example:

None

1 million 1KB files = huge metadata pressure

Better:

None

Merge files

Use Parquet/ORC

Use batching

15. Common Tuning Parameters

None

Replication factor = 3

Block size = 128MB / 256MB

Heartbeat interval

Rack awareness config

16. Advantages

None

Cheap horizontal scaling

Fault tolerant

Good for TB/PB data

Parallel processing

Excellent with Spark/Hadoop ecosystem

17. Limitations

None

Not low-latency OLTP

Poor for small files

Not ideal for random updates

NameNode complexity

High network/storage overhead from replicas

18. Used With

- Apache Spark
- Apache Hive
- Apache MapReduce
- Apache HBase

19. Interview HLD Talking Points

If suggesting HDFS, mention:

None

Batch workloads

Data lake

Immutable logs

Replication factor

Rack awareness

Small file mitigation

HA NameNode

20. One-Line Summary

HDFS replication stores each file block on multiple DataNodes across racks, enabling fault-tolerant, scalable storage for large-scale analytics workloads.